Real-Time Driver Distraction Detection: A 2D vs. 3D Vision Benchmark

Team Members:

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1. Problem Description

Driver inattention, caused by drowsiness, phone usage, eating, or other distractions, remains one of the leading causes of traffic accidents worldwide. Developing an intelligent, computer-vision-based driver monitoring system (DMS) can significantly reduce accident rates by providing real-time alerts.

However, real-time driver action recognition presents distinct technical challenges:

1. **Spatial Complexity:** Recognizing subtle hand-eye-object interactions (e.g., holding a mobile phone vs. holding the steering wheel).
2. **Temporal Dynamics:** Distinguishing continuous temporal actions (e.g., nodding off/yawning) from static postures.
3. **Latency vs. Accuracy Trade-Off:** Deploying deep neural networks in vehicle cabins on edge devices requires low inference latency without sacrificing accuracy.

This project addresses the problem of driver distraction classification by comparing single-frame spatial feature representations against multi-stream temporal networks.

2. Literature Survey & Proposed Work

2.1 Literature Survey

Prior works in driver distraction detection generally fall into two architectural paradigms:

1. **Frame-Level Spatial CNNs:** Early approaches (e.g., Baheti et al., CVPRW 2018; Abouelnaga et al., CVPRW 2018) process individual video frames using standard Convolutional Neural Networks (e.g., VGG, ResNet) to classify driver postures. While computationally lightweight and simple to implement, these methods lack temporal context, making them prone to false positives during rapid pose transitions

2. **Spatio-Temporal Video Architectures:** Fundamental 3D convolutional networks (Tran et al., ICCV 2015) and recent advances such as SlowFast Networks (Feichtenhofer et al., ICCV 2019) utilize multi-pathway 3D-CNNs - a **Slow pathway** operating at low frame rates to capture spatial semantics, and a *Fast pathway* operating at high refresh rates to capture motion patterns.

2.2 How Proposed Work is Different

While existing studies evaluate spatial models and video networks separately on isolated datasets, our proposed work provides a direct benchmark comparison between **Frame-Level CNNs** and **3D Spatio-Temporal SlowFast Networks** on the same driver monitoring dataset.
Specifically, we propose to:

- Evaluate the tradeoff between **Inference Latency (FPS)**, **Model Size (MB)**, and **Classification Accuracy/F1-Score**.
- Analyze failure modes where frame-level models fail due to a lack of temporal cues (e.g., distinguishing a quick glance at the mirror from prolonged distraction).

3. Preliminary Plan

1. **Data Preparation & Pipeline Setup**
    - Download and set up the driver dataset (State Farm / AUC Distracted Driver)
    - Build data loading and preprocessing pipelines for both single-frame and temporal video inputs
2. **Model Implementation & Initial Training**
    - **An Dang:** Implement/fine-tune the baseline Frame-Level CNN model (Baheti et al.)
    - **Huy Pham:** Implement/fine-tune the dual-pathway SlowFast network (Feichtenhofer et al.)
3. **Intermediate Report**
    - Analyze initial validation performance and training metrics
    - Submit Intermediate Project Report
4. **Experimental Benchmarking & Analysis**
    - Run direct comparisons on accuracy, F1-score, model size, and inference latency (FPS)
    - Perform failure mode and error analysis on edge cases
5. **Final Presentation**
    - Prepare presentation slides and live demo
    - Deliver 5-8 minute project presentations
6. **Final Submission**
    - Clean up GitHub repository with structured commits and documentation
    - Complete and submit the Final Project Report in LaTeX CVPR template

4. References

1. Baheti, B., Gajre, S., & Talbar, S. (2018). Detection of Distracted Driver Using Convolutional Neural Network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1032-1038.

2. Abouelnaga, Y., Eraqi, H. M., & Moustafa, M. N. (2018). Real-time Distracted Driver Posture Classification. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1040-1048.

3. Ma, Y., & Wang, Z. (2022). ViT-DD: Multi-Task Vision Transformer for Semi-Supervised Driver Distraction Detection. arXiv. https://doi.org/10.48550/arxiv.2209.09178

4. Feichtenhofer, C., Fan,https://doi.org/10.48550/arxiv.2209.09178 H., Malik, J., & He, K. (2019). SlowFast Networks for Video Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6202-6211.
   Github link: https://github.com/facebookresearch/SlowFast

5. Tran, D., Bourdev, L., Fergus, R., Torresani, L., & Paluri, M. (2015). Learning Spatiotemporal Features with 3D Convolutional Networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 4489-4497.